

Predictive Coding Tendencies in the GPT-2 Family: Multi-Method Empirical Evidence Across Five Independent Experiments

Jong-O Yun
Independent Researcher, South Korea
`gallam.research@gmail.com`

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Abstract

Whether transformer-based language models operate according to predictive coding (PC) principles remains unclear. In this paper, I investigate empirical evidence for predictive coding tendencies in transformers through five independent experiments, using six models centered on the GPT-2 family, with GPT-Neo 125M and Pythia 160M/410M as architectural controls.

In the GPT-2 family, English showed the most stable residual stream convergence across layers. I found that the W-shaped MLP magnitude pattern is consistent across GPT-2 models regardless of size, and that removing the convergence zone caused the largest performance drop — with all 95% confidence intervals excluding zero ($[0.25, 0.44]$, $[0.36, 0.75]$, $[0.53, 1.15]$). The logit lens confirmed that rank reduction during the early spike zone is 17–110 $\times$ faster than during the convergence zone. However, these patterns were not consistently observed in the control models GPT-Neo and Pythia, suggesting dependence on training configuration rather than transformer architecture in general.

These predictive coding tendencies appear consistently across multiple independent methods, but do not constitute mechanistic proof.

1 Introduction

Transformer-based language models demonstrate remarkable capabilities, yet their internal computational mechanisms remain incompletely understood. A fundamental question shared by neuroscience and artificial intelligence is whether these models implement principles analogous to those observed in biological neural systems.

Predictive coding (PC) [Rao and Ballard, 1999] is a neuroscience theory proposing that the brain continuously generates hierarchical predictions and minimizes prediction errors. Millidge et al. [2022] proved that under certain conditions, predictive coding is mathematically equivalent to backpropagation, suggesting that transformers trained via backpropagation may implicitly perform computations similar to predictive coding.

I present multi-method empirical evidence using the GPT-2 family as the primary subject of analysis, with GPT-Neo 125M and Pythia 160M/410M as architectural controls. Key contributions:

1. **GPT-2 family residual stream analysis** across seven languages, revealing that convergence patterns depend on training configuration rather than architecture alone.
2. **Zero-ablation** demonstrating that the low-magnitude convergence zone is functionally more critical than spike zones in GPT-2 models—consistent with PC’s error-minimization phase.
3. **Logit lens quantification** of monotonic prediction refinement in GPT-2 models, providing evidence of hierarchical representation convergence consistent with (though not proof of) PC.
4. **Multi-method convergence:** activation patching, zero-ablation, and logit lens independently identify mid-layer processing as the most decisive region across all GPT-2 variants.

2 Related Work

2.1 Predictive Coding in Neuroscience and Deep Learning

Rao and Ballard [1999] proposed that the brain minimizes prediction errors via hierarchical top-down predictions and bottom-up error signals. Millidge et al. [2022] proved that PC is mathematically equivalent to backpropagation, implying that backpropagation-trained networks may implicitly implement PC-like computations. Salvatori et al. [2022] extended PC-based learning rules to diverse architectures.

2.2 Mechanistic Interpretability

Elhage et al. [2021] introduced the residual stream framework for principled analysis of information flow in transformers. Meng et al. [2022] showed via activation patching that factual associations are localized to specific MLP layers—a methodology I adopt in Experiment 2. Cunningham et al. [2023] and Olah et al. [2020] developed sparse autoencoders and circuit analysis for interpreting transformer internals.

2.3 Layer Depth and Residual Stream Analysis

nostalgebraist [2020] introduced the logit lens, projecting intermediate residual streams to vocabulary space—which I use directly in Experiment 5. Hu et al. [2025] introduced “effective depth,” finding that early layers drive feature composition while later layers perform minor refinements, consistent with the W-shaped MLP magnitude pattern I observe.

2.4 Cosine Similarity as Representational Stability

Hinostroza et al. [2026] showed cosine similarity is a poor proxy for layer importance. I use it as a measure of *representational stability* rather than relevance, consistent with PC’s error-minimization framing. Experiment 4 provides independent causal validation.

2.5 Attention Sink Phenomenon

Xiao et al. [2023] identified attention sinks at boundary tokens, a phenomenon that bears structural resemblance to PC’s boundary reset signals.

3 Methods

3.1 Models

I conducted all experiments using TransformerLens [Nanda and Bloom, 2022] and Google Colab’s T4 GPU. The primary analysis targets are GPT-2 Small, Medium, and Large, with GPT-Neo 125M, Pythia 160M, and Pythia 410M serving as architectural controls.

Table 1: Model configurations used in this study.

Model	Params	Layers	Training Data	Tokenizer
GPT-2 Small	117M	12	WebText	GPT-2 BPE
GPT-2 Medium	345M	24	WebText	GPT-2 BPE
GPT-2 Large	774M	36	WebText	GPT-2 BPE
GPT-Neo 125M	125M	12	The Pile [Gao et al., 2021]	GPT-2 BPE
Pythia 160M	160M	12	The Pile	NeoX
Pythia 410M	410M	24	The Pile	NeoX

All models were loaded with frozen weights.

3.2 Experiment 1: Residual Stream Convergence

I measured layer-wise cosine similarity and delta magnitude across 140 prompts in seven languages: English (30), Korean (20), Spanish (20), French (20), Russian (20), Japanese (20), and Arabic (10).

For layer transition $l \rightarrow l + 1$:

$$\text{sim}(l) = \frac{1}{T} \sum_{t=1}^T \cos(\mathbf{r}_l^t, \mathbf{r}_{l+1}^t), \quad \delta_l = \|\mathbf{r}_{l+1} - \mathbf{r}_l\|_2. \quad (1)$$

Statistical significance was assessed via two-sample t -tests comparing early layers (first 40%) against late layers (last 40%).

3.3 Experiment 2: Activation Patching

I measured causal activation effects by patching 23 semantic pairs (capital cities (10), currencies (5), language relations (5)) across layers and token positions. For each semantic pair (clean x_c , corrupt x_p):

$$\text{Effect}(l, t) = \text{logit}_{\text{patched}(l, t)}(\text{target}) - \text{logit}_{\text{patched}(l, t)}(\text{foil}), \quad (2)$$

where $\text{patched}(l, t)$ substitutes activations from x_c at layer l , position t into the corrupt run [Meng et al., 2022]. For cross-model comparison, I additionally report the average $\text{Effect}(l, t)$, normalized per pair to $[0, 1]$, over subject-token positions (token indices 2–3) and mid-layer range $0.25 \leq l/L \leq 0.33$.

3.4 Experiment 3: MLP Transform Magnitude

I measured how much each MLP layer transforms its input using the L2 norm of the output minus input:

$$\mu_l = \|\text{MLP}_l(\mathbf{x}) - \mathbf{x}\|_2. \quad (3)$$

Layer indices were normalized to $[0, 1]$ for cross-model comparison.

3.5 Experiment 4: Zero-Ablation

I divided layers into three zones and measured normalized performance drop when all MLP outputs in a zone are zeroed:

$$\text{drop}(z) = \frac{\text{logitdiff}_{\text{base}} - \text{logitdiff}_{\text{ablated}(z)}}{|\text{logitdiff}_{\text{base}}|}. \quad (4)$$

Table 2: MLP zone definitions.

Zone	Norm. Position	Hypothesized PC Role
Early Spike	0.10–0.20	Initial error signal processing
Convergence	0.25–0.75	Error minimization
Late Spike	0.80–0.95	Final prediction output

I used 37 semantic pairs with single-token answers, retaining only pairs with $\text{logitdiff}_{\text{base}} > 1.0$ ($n = 25$ –28 per GPT-2 model).

3.6 Experiment 5: Logit Lens

I used the logit lens [nostalgebraist, 2020] to track the rank of the correct next token at each layer:

$$\text{rank}_l = \text{rank}(\mathbf{r}_l \cdot W_U + b_U, y), \quad (5)$$

averaged over 15 English prompts per model.

4 Results

4.1 Finding 1: Training-Configuration-Dependent Convergence

Table 3: Mid-layer cosine similarity (Sim) and delta p -values (p_δ). Bold = highest per model. En rank = English rank among 7 languages.

Lang.	GPT-2 S		GPT-2 M		GPT-2 L		GPT-Neo		Pythia 160M		Pythia 410M	
	Sim	p_δ	Sim	p_δ	Sim	p_δ	Sim	p_δ	Sim	p_δ	Sim	p_δ
en	.972	.406	.989	.086	.988	.008	.982	.799	.896	.031	.937	.465
es	.958	.043	.985	<.001	.984	<.001	.982	.866	.848	.001	.929	.337
fr	.957	.065	.985	.001	.984	<.001	.982	.921	.854	.002	.929	.388
ko	.962	<.001	.986	<.001	.984	<.001	.982	.066	.873	<.001	.941	.345
ru	.961	<.001	.986	<.001	.984	<.001	.981	.026	.874	<.001	.936	.407
ja	.968	<.001	.988	.001	.986	<.001	.983	.352	.896	<.001	.943	.513
ar	.966	.003	.988	<.001	.986	<.001	.983	.074	.879	<.001	.938	.473
En rank	#1		#1		#1		#3		#1		#4	

In the GPT-2 family, English showed non-significant delta differences between early and late layers — the English residual stream remained uniformly stable across layers. Non-English languages showed significant early-to-late delta differences. Even in control models, English ranked differently: 3rd in GPT-Neo (0.982) and 4th in Pythia 410M (0.937). Since GPT-Neo shares the same BPE tokenizer as GPT-2, tokenizer differences alone cannot explain this divergence, confirming that training data composition and learning procedure are the determining factors.

After Bonferroni correction ($\alpha = 0.05/42 \approx 0.0012$), English remains non-significant in all three GPT-2 models ($p \geq 0.008$), preserving the core finding.

I use cosine similarity as a measure of *representational stability* rather than layer importance [Hinostroza et al., 2026]. Experiment 4 provides independent causal validation that does not rely on this metric.

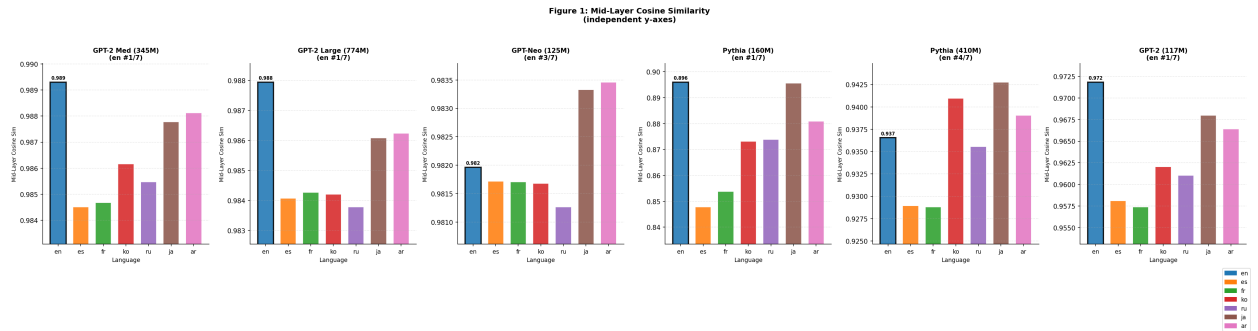


Figure 1: Mid-layer cosine similarity by language across six models (independent y -axes). English rank shown in parentheses.

4.2 Finding 2: Mid-Layer Semantic Processing

Across all six models and 23 semantic pairs, semantic tokens showed peak activation patching effects at normalized layer position 0.25–0.33, while function words showed minimal effects throughout. The average mid-layer subject-token patching effect was: GPT-2 Small = 0.112, Medium = 0.174, Large = 0.143, GPT-Neo = 0.206, Pythia 160M = 0.168, Pythia 410M = 0.146. The total range of 0.11–0.21 shows quantitatively consistent mid-layer semantic processing across all six models.

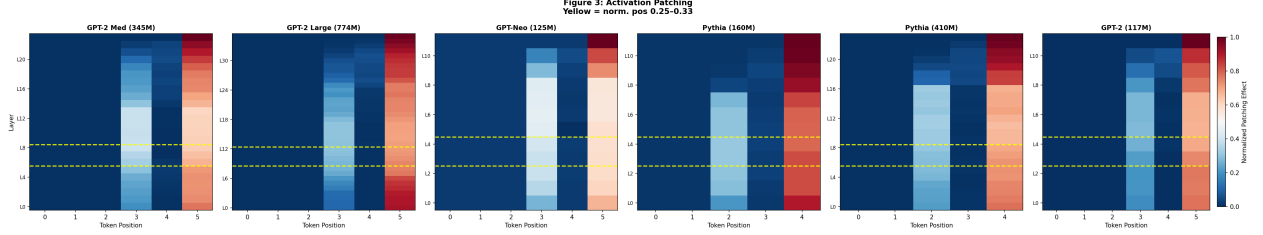


Figure 2: Activation patching effect across layers and token positions. Yellow dashed lines: normalized positions 0.25–0.33.

4.3 Finding 3: W-Shaped MLP Magnitude Pattern

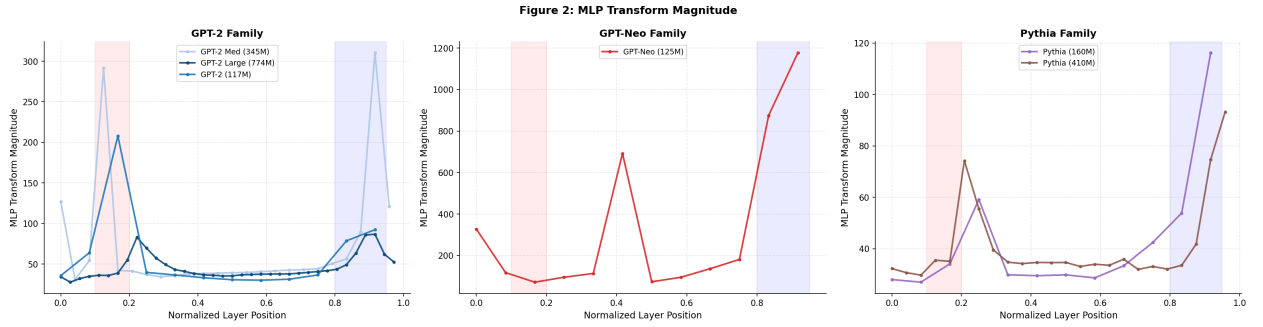


Figure 3: MLP transform magnitude by model family. GPT-2 (left) shows a clear W-shape; GPT-Neo (center) shows an additional mid-section spike; Pythia (right) shows early and late elevation with a less pronounced convergence trough.

In the GPT-2 family, regardless of the number of layers, a consistent three-zone structure emerged at normalized layer positions: early spike (0.10–0.20; Layer 2 Small: 409.4, Layer 3 Medium: 578.7, Layer 8 Large: 133.5), convergence (0.20–0.75; 27–50 units), and late spike (0.85–0.95). This W-shaped pattern is scale-invariant, consistent with Hu et al. [2025]. GPT-Neo shows a significant additional spike within the convergence zone, while Pythia shows a less pronounced convergence trough.

4.4 Finding 4: Functional Primacy of the Convergence Zone

Table 4: Zero-ablation results: normalized performance drop (mean, 95% CI, n = valid pairs). Bold = largest drop per model. All GPT-2 Convergence zone 95% CIs exclude zero.

Zone	GPT-2 Small	GPT-2 Medium	GPT-2 Large
Early Spike	+0.10 [0.01, 0.19] ($n=25$)	+0.18 [0.06, 0.30] ($n=26$)	+0.05 [−0.01, 0.11] ($n=28$)
Convergence	+0.35 [0.25, 0.44] ($n=25$)	+0.56 [0.36, 0.75] ($n=26$)	+0.84 [0.53, 1.15] ($n=28$)
Late Spike	+0.00 [−0.07, 0.08] ($n=25$)	+0.03 [−0.04, 0.10] ($n=26$)	+0.09 [−0.01, 0.19] ($n=28$)

When the convergence zone — which has the lowest MLP transform magnitude — was removed, performance dropped the most. All 95% CIs for the convergence zone exclude zero, statistically confirming the performance drop. Early Spike and Late Spike CIs include or approach zero, indicating weaker effects. The functional importance of the convergence zone increases with model scale (mean drops: 0.35, 0.56, 0.84).

GPT-Neo shows the opposite pattern, with the late spike zone (+0.85, CI [0.59, 1.11]) outweighing the convergence zone (+0.54, CI [0.15, 0.92]). Pythia 160M and 410M also show convergence zone primacy (+0.61, CI [0.30, 0.91] and +1.09, CI [0.64, 1.54] respectively), though Pythia 160M’s late spike zone yielded an extreme outlier (−2.64, clipped in Figure 4) that warrants caution.

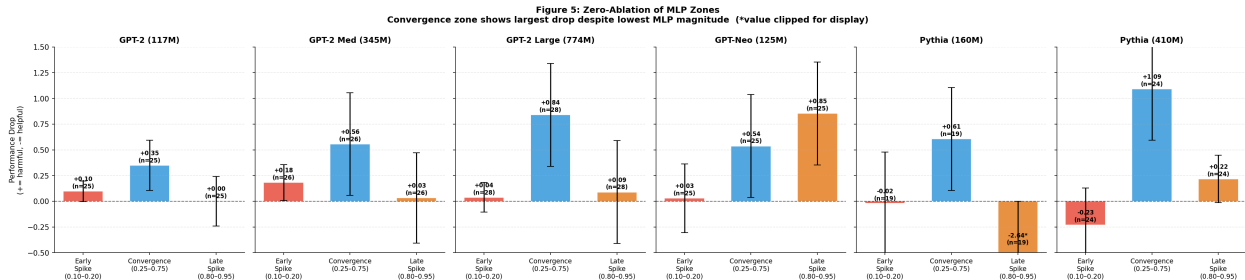


Figure 4: Zero-ablation results by MLP zone across all six models (y-axis clipped to $[-0.5, 1.5]$; asterisk indicates values outside range). The Convergence zone (blue) shows the largest performance drop in all GPT-2 models. GPT-Neo shows a different pattern where the Late Spike zone dominates.

4.5 Finding 5: Monotonic Prediction Refinement

Table 5: Average target token rank by layer zone for GPT-2 models (lower = better), and rank reduction rate (rank drop ÷ normalized zone width). GPT-Neo and Pythia discussed separately.

Model	Avg Rank				Rate (rank/norm-unit)		
	Layer 0	After Early	After Conv	Final	Early Spike	Convergence	Late Spike
GPT-2 Small (12L)	374.5	93.8	2.5	0.53	2,807	164	13
GPT-2 Medium (24L)	275.8	25.5	2.8	0.47	2,503	41	10
GPT-2 Large (36L)	5,591.5	268.4	2.1	0.67	53,231	484	7

In all three GPT-2 models, the target token rank decreased monotonically from Layer 0 to the final layer: Small $375 \rightarrow 0.53$, Medium $276 \rightarrow 0.47$, Large $5,592 \rightarrow 0.67$. The rate of rank reduction was strongly concentrated in the early spike zone: $17\times$, $61\times$, and $110\times$ faster than the convergence zone respectively, with 75–95% of total rank reduction occurring during the early spike zone.

This rate profile is consistent with predictive coding’s hierarchical prediction refinement, but is not unique evidence for PC — monotonic rank reduction is expected in any well-trained language model.

GPT-Neo showed a non-monotonic pattern where rank initially increased during the early spike zone before decreasing. Pythia 160M showed minimal rank change during the early spike zone, with a sharp drop occurring during the convergence zone instead. Pythia 410M showed a pattern similar to GPT-2, with rapid early reduction reaching a final rank of 0.40.

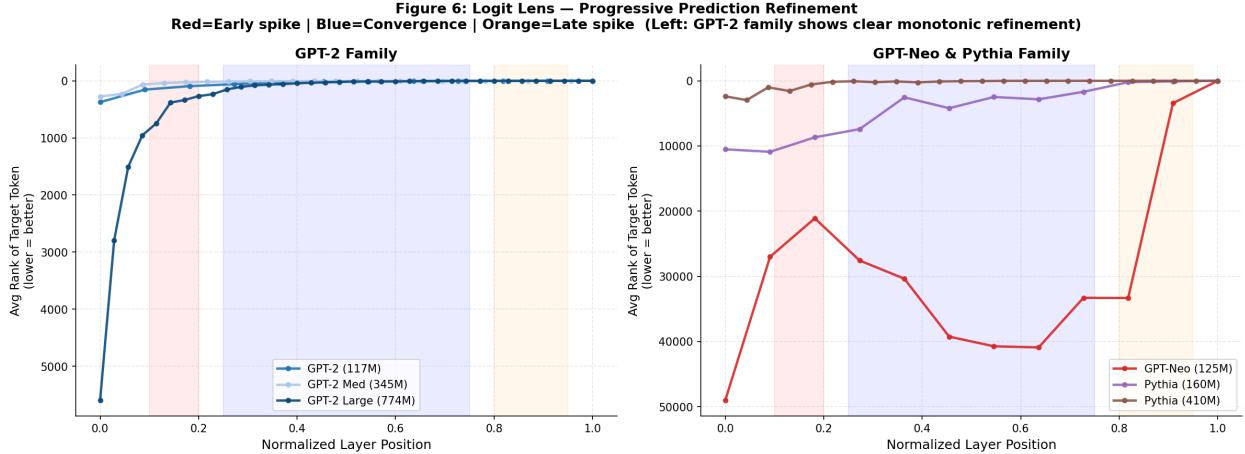


Figure 5: Logit lens: average rank of target token vs. normalized layer position (y -axis inverted; lower = better). *Left*: GPT-2 family shows clear monotonic rank reduction. *Right*: GPT-Neo and Pythia show qualitatively different patterns. Shaded regions: Early Spike (red), Convergence (blue), Late Spike (orange).

5 Discussion

5.1 Convergence of Three Independent Methods

In the GPT-2 family, three independent methods — activation patching, zero-ablation, and logit lens — all consistently point to the mid-layer convergence zone as the most functionally critical region. This is difficult to attribute to coincidence, and I believe this finding has sufficient value to motivate further research.

5.2 Dissociation of Magnitude and Functional Importance

In predictive coding, the early and late zones can be thought of as transmission points for prediction signals and error signals respectively. I initially expected that the convergence zone — where these two signals meet — would show the highest MLP transform magnitude. The actual results showed the opposite: MLP transform magnitude in the convergence zone was the lowest at 27–50 units, yet removing it caused the largest performance drop (GPT-2 Small [0.25, 0.44], Medium [0.36, 0.75], Large [0.53, 1.15]), with all confidence intervals excluding zero. I interpret this as the convergence zone performing precise, small-scale error minimization rather than large-scale transformation — and that these small adjustments are the most consequential for correct prediction.

5.3 Cross-Architecture Variation

Even among transformer-based language models, the strength of PC tendencies varied substantially depending on training configuration. The GPT-2 family showed consistent PC tendencies across all five experiments — English convergence ranked first, W-shaped MLP pattern, convergence zone 95% CIs all excluding zero, and monotonic logit lens rank reduction. In contrast, GPT-Neo’s English ranked 3rd in convergence and its late spike zone outweighed the convergence zone in zero-ablation. Pythia showed non-monotonic patterns in the logit lens. Notably, activation patching showed similar mid-layer effects across all six models (range: 0.11–0.21), suggesting that mid-layer semantic processing may be more architecturally robust than the convergence patterns observed in Experiment 1.

5.4 Limitations

The limitations of these experiments are: first, there is no mechanistic proof at the circuit level that transformers implement predictive coding; and second, I was unable to determine the reason for the differences

between the GPT-2 family and the control models GPT-Neo and Pythia. To establish this, future work is needed that analyzes what computations actually occur at the circuit level. Additional limitations include: high inter-pair variance in zero-ablation (GPT-2 Large Convergence CI: [0.53, 1.15]); potential tokenizer confounds in cross-language cosine similarity comparisons; and semantic pairs restricted primarily to encyclopedic facts.

6 Conclusion

In conclusion, I observed computational tendencies consistent with predictive coding in GPT-2 family models, supported by five independent experiments including the W-shaped MLP magnitude pattern and zero-ablation results. Since the same patterns were not consistently observed in the control models GPT-Neo and Pythia, these tendencies appear to depend on the specific training configuration of the GPT-2 family rather than being a universal property of transformer-based models.

Key findings:

1. **Training-configuration-dependent convergence:** English ranks first in all GPT-2 models, strengthening with scale; this pattern weakens in GPT-Neo and Pythia.
2. **Mid-layer semantic specialization:** Activation patching identifies normalized position 0.25–0.33 as the peak causal locus, consistent across all six models (range: 0.11–0.21).
3. **W-shaped MLP magnitude:** A spike–convergence–spike structure in GPT-2 is scale-invariant; a dual-spike structure appears in GPT-Neo and Pythia.
4. **Convergence zone functional primacy:** Zero-ablation shows the low-magnitude convergence zone is most critical, with importance growing with scale and all 95% CIs excluding zero.
5. **Monotonic prediction refinement:** Logit lens confirms hierarchical rank reduction in all GPT-2 variants; the rate profile is consistent with (though not uniquely predicted by) PC.

These findings represent multi-method empirical evidence of tendencies, not mechanistic proof. Future mechanistic work should validate these tendencies at the circuit level, extend to additional model families with controlled training configurations, and develop PC-specific predictions that can be distinguished from alternative explanations.

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A Supplementary Figures

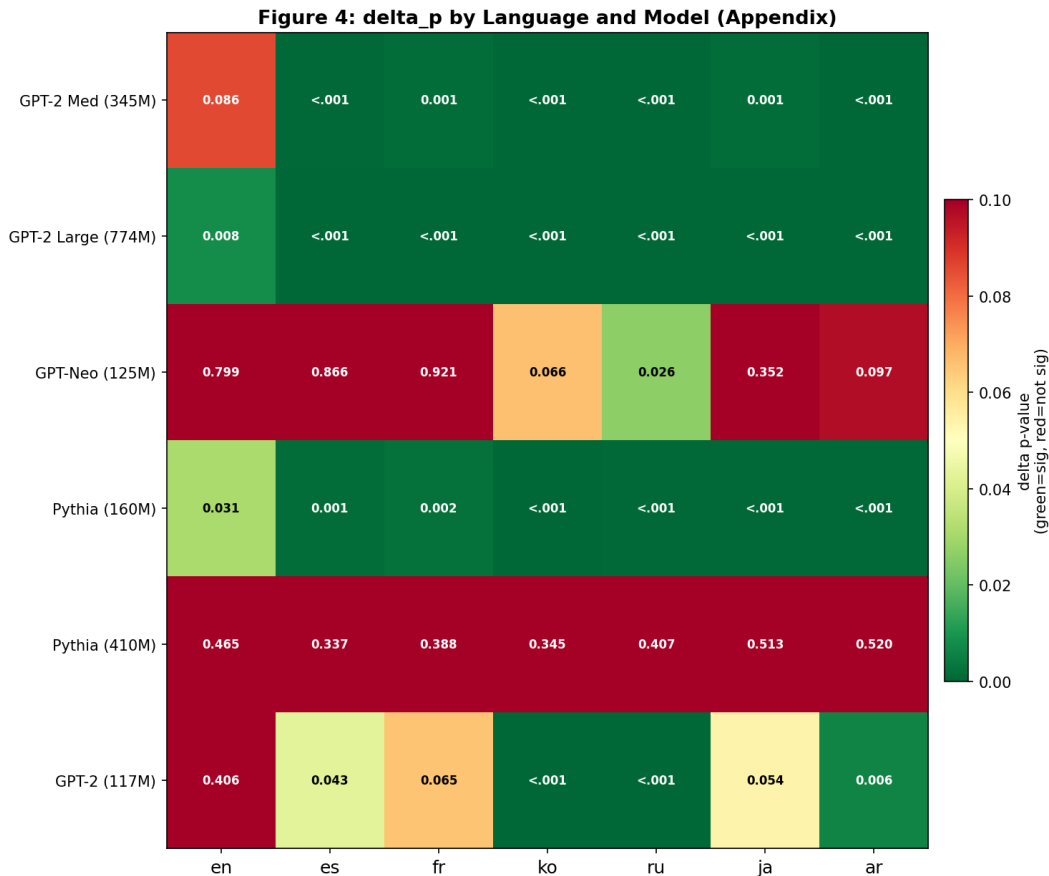


Figure 6: Delta p -values (p_δ) by language and model. Each cell shows the p -value of a two-sample t -test comparing early-layer (first 40%) vs. late-layer (last 40%) residual stream delta magnitudes. Green = significant ($p < 0.05$); red = not significant. The GPT-2 family shows English as the most stable language. GPT-Neo shows no clear language-dependent pattern. Pythia 410M shows no significant patterns across any language. After Bonferroni correction ($\alpha = 0.0012$), English remains non-significant in all GPT-2 models.